

Ja-Chan Lu

US Citizen | jachanlu056@gmail.com | (213) 362 8058 | San Diego, California | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of California, San Diego

B.S. in Data Science, Rising Junior

La Jolla, California

Expected Graduation, June 2028

- **Relevant CS Coursework:** Introduction/Principles/Practice of Data Science, Data Structures and Algorithms, Theoretical Foundations I and II of Data Science, Machine Learning Algorithms, Data Management, Econometrics, Linear Algebra, Calculus

PROJECT EXPERIENCE

NBA Player Performance — Predictive Modeling

- Quantified “stretch big” archetype by discovering 1,906% surge in 3-point attempts among bigs from 1996-2024, outpacing even Guards and Wings, signifying the shift in structure of modern NBA offense.
- Led a 4 person team to ingest/clean 6,000 player-seasons across 3 historical eras using NBA API and building reproducible python pipelines that shortened manual data-cleaning time by 10+ hours.
- Modeled shifting mathematical relationship between physical traits and 3-point efficiency, isolating height as significant negative predictor across all 3 tested eras.

Multi-Sector Equity Crash Study

- Engineered automated financial pipeline in R utilizing tidyquant to calculate max drawdown, crash beta, recovery time for 47 equities across 6 market sectors.
- Isolated historical data from 2019-2023 to compare performance across 2 distinct crashes (COVID demand shock and 2022 inflation bear market) and test sector behavior across various scenarios
- Identified Healthcare sector as the most consistent protector during downturns, whereas Energy/Tech sector experience total performance reversals between demand-driven and inflation-driven regimes.

NBA Veteran Efficiency — SQLite Ranking Engine

- Designed and built multi-table SQLite database to store, index, and query 500+ player performance statistics
- Wrote complex SQL queries with aggregate windows and multi-table joins to compute custom per-minute efficiency stats benchmarked against league-wide averages
- Found that versatility and court vision define veteran success, not athleticism

League of Legends — Carry Potential ML Pipeline

- Analyzed 117,000+ professional match records using Oracle’s Elixir with EDA, permutation hypothesis testing, and predictive Decision Tree Classifier.
- Built/deployed pipeline as interactive web report, achieving 90% model accuracy.
- Determined 15-minute gold differential as single strongest match outcome predictor, completely ignoring strength of team compositions and revealing importance of drafting strong, champions that excel earlier on in the match

Work Experience

Chinatown Service Center

Intern Tutor (Paid)

Los Angeles, California

October 2022 – June 2023

- Maintained perfect attendance and punctuality across 9 months, reliably leading cohort of 6th graders in multiple 3+ hour sessions
- Tracked students of all ages’ progress to identify learning gaps and adjusted instructions, contributing to measurable improvements in test scores and participation
- Communicated complex concepts clearly to ESL students and families in Cantonese, Mandarin, English

SKILLS & INTERESTS

- **Languages & Tools:** Python, SQL, Java, R, Git, GitHub, Jupyter
- **Frameworks & Libraries:** PyTorch, Pandas, NumPy, scikit-learn, tidyverse, ggplot2, tidyquant
- **Techniques:** Financial Analysis, Data Analysis, Database Design, Data Aggregation, Machine Learning, Data Visualization, Hypothesis Testing, Time Series, EDA, MLOps,
- **Interests:** Basketball, Piano, Weightlifting, Cafe-Visiting
- **Languages Spoken:** English, Cantonese, Mandarin